

Supplemental meterial

I. SECURITY ANALYSIS

We analyze DAST and DAST+ under the semi-honest and adaptive adversarial model defined in Sec.III-C. The cloud server follows the protocol specification but may try to infer information from the encrypted index, query tokens, response sizes, and the observed access and volume patterns. The adversary may also choose later queries adaptively based on the leakage obtained from previous queries. Our goal is to show that, assuming the underlying RSSE primitive is adaptively IND-CKA2 secure, the proposed constructions reveal no information beyond the stated leakage functions. We do not claim forward or backward privacy for dynamic updates.

Leakage Functions. Let MM be the plaintext multimap materialized by the index construction, and let Σ be the encrypted multimap obtained from the underlying RSSE scheme. The setup leakage is defined as

$$\mathcal{L}_{\text{setup}}(\text{DB}) = (\mathcal{L}_{\text{setup}}^{\Sigma}(\text{MM}), |\mathcal{A}|, \{\text{TNodes}_j\}_{A_j \in \mathcal{A}}),$$

which includes the leakage of the underlying encrypted multimap and public size information of the adaptive temporal forest.

For a query $q = (R_t, R_s, \varphi_{\text{key}}, \varphi_{\text{join}})$, the query leakage is defined as

$$\mathcal{L}_{\text{query}}(\text{DB}, q) = (\text{Str}(\text{DB}, q), \{\mathcal{L}_{\text{query}}^{\Sigma}(\text{MM}, w)\}_{w \in W(q)}),$$

where $W(q)$ is the set of block-local range keys issued by the query. The structural pattern $\text{Str}(\text{DB}, q)$ reveals how the query covers the range-supporting structure, including the adaptive temporal blocks touched by R_t , the block-local covered nodes selected by the segment tree or TDAG cover, and the corresponding response volumes. It does not reveal the plaintext records, spatial coordinates, keywords, or join values beyond the controlled leakage of the underlying RSSE scheme.

Theorem 1. *Let Σ be an adaptively IND-CKA2-secure RSSE scheme with leakage $(\mathcal{L}_{\text{setup}}^{\Sigma}, \mathcal{L}_{\text{query}}^{\Sigma})$. Then any construction instantiated from Σ and a range-supporting structure $(\mathcal{I}, \text{RC})$, including DAST and DAST+, is adaptively IND-CKA2 secure with respect to $(\mathcal{L}_{\text{setup}}, \mathcal{L}_{\text{query}})$.*

Proof: We construct a stateful simulator $\mathcal{S}$ that uses only $(\mathcal{L}_{\text{setup}}, \mathcal{L}_{\text{query}})$ to simulate the view of any PPT adversary.

Setup. In the real execution, the client first constructs MM from DB and the range-supporting structure, and then runs RSSE.Setup to obtain the encrypted multimap Σ . In the ideal execution, $\mathcal{S}$ receives $\mathcal{L}_{\text{setup}}(\text{DB})$, including $\mathcal{L}_{\text{setup}}^{\Sigma}(\text{MM})$ and the public size of the adaptive temporal forest. It invokes the

setup simulator of the underlying RSSE scheme to generate a simulated encrypted multimap $\tilde{\Sigma}$ with the same leaked size and format. By the adaptive security of Σ , $\tilde{\Sigma}$ is computationally indistinguishable from the real encrypted index.

Queries. Consider an adaptive query sequence $q^{(1)}, \dots, q^{(t)}$. For each query $q^{(i)}$, the real client computes a range cover over the adaptive temporal forest. For DAST, this cover consists of the segment-tree nodes selected inside each query-touched adaptive temporal block. For DAST+, it consists of the TDAG node, or a constant number of TDAG nodes, selected inside each touched block. The client then generates RSSE tokens for the corresponding range keys and obtains the encrypted responses from the server.

In the ideal execution, $\mathcal{S}$ receives only $\mathcal{L}_{\text{query}}(\text{DB}, q^{(i)})$. From $\text{Str}(\text{DB}, q^{(i)})$, it learns the number of touched adaptive temporal blocks, the selected block-local cover nodes, and the response volumes. For each leaked range key $w \in W(q^{(i)})$, it invokes the query simulator of the underlying RSSE scheme on $\mathcal{L}_{\text{query}}^{\Sigma}(\text{MM}, w)$ to obtain a simulated token and simulated response. The simulated outputs are returned to the adversary in the same order and with the same leaked volume as in the real execution.

Hybrid argument. Let Hyb_0 be the real execution. Let Hyb_1 be identical to Hyb_0 except that the encrypted multimap, query tokens, and responses generated by Σ are replaced by the outputs of its setup and query simulators. By the adaptive IND-CKA2 security of Σ , no PPT adversary can distinguish Hyb_0 from Hyb_1 except with negligible probability. Since Hyb_1 is exactly the ideal execution defined by $\mathcal{S}$ and the leakage functions above, the theorem follows. ■

Leakage Analysis of DAST. DAST exhibits **localized structural leakage** through *token co-occurrence*. Since a segment-tree range cover may emit multiple block-local tokens for each query-touched adaptive temporal block, the server can observe fine-grained co-occurrence patterns among tokens. A persistent adversary may exploit these patterns to infer partial block-local temporal adjacency or cover relations, thereby learning more about the local segment-tree decomposition.

Leakage Analysis of DAST+. DAST+ reduces fine-grained co-occurrence leakage by using a TDAG-based range cover. For a query touching d adaptive temporal blocks, it emits $O(d)$ tokens, i.e., one or a constant number of tokens per touched block. By mapping a block-local range to a coarser TDAG node and allowing controlled temporal false positives, DAST+ hides the fine-grained segment-tree decomposition and increases the uncertainty of co-occurrence-based reconstruction attacks.

To sum up, both DAST and DAST+ are IND-CKA2-

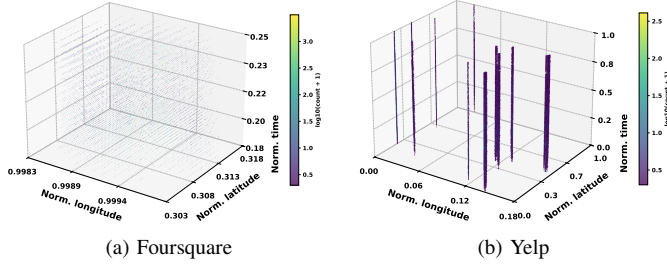


Fig. 1: Spatiotemporal distributions of Foursquare and Yelp datasets.

secure under well-defined leakage functions. **DAST** provides higher query precision but exposes a richer localized structural pattern, whereas **DAST+** trades controlled false positives for a smaller fine-grained token co-occurrence surface.

II. PERFORMANCE EVALUATION

A. Experimental Setup

Datasets. To evaluate the proposed schemes under heterogeneous real-world spatiotemporal distributions, we use two public check-in datasets with clearly different spatial, temporal, and joint spatiotemporal characteristics.

- **Foursquare:** This dataset contains approximately 33 million check-in records globally from 2008 to 2013. As shown in Fig. 1a, for our evaluation, we focus on the spatiotemporal trajectories to assess performance in non-uniform, high-density environments.
- **Yelp:** This dataset contains timestamped location records associated with user activities and business locations. As shown in Fig. 1b, Yelp presents multiple separated spatial clusters over a broader normalized temporal range, forming several vertical spatiotemporal bands.